Title	Authors	Abstract	EXTRA
GIS-Based Spatio-Temporal Analysis of Vegetation Dynamics, Urban Expansion, and Land Surface Temperature in Davao City	Richelyn C. DumdumJason Ben R. Paragamac	**Abstract:** The Earth's surface and atmosphere have been substantially modified by the growth of anthropogenic activities and urbanization. Changes in the composition and operation of the urban ecological system will affect the health of city dwellers by increasing the risk of heat exposure, increasing the risk of violence, illness, and mortality, and reducing labor productivity. Davao City is currently undergoing rapid urbanization, and no comprehensive study has examined the effects of the changes on vegetation indices, urban expansion, traffic density, and land surface temperature, which are likely to affect the area. This study analyzes the spatio-temporal change in vegetation indices, built-up expansion, traffic density, and land surface temperature from 2018 to 2022 using Landsat 8 imagery and GIS tools, particularly QGIS. The results show a moderate negative relationship between NDBI and NDVI (r = −0.4313), a strong positive relationship between NDBI and LST (r = 0.6216), and a weak negative relationship between NDVI and LST (r =−0.2495). Results highlight the interconnected influence of urbanization, vegetation dynamics, and thermal conditions within the study area, underscoring the need to integrate green infrastructure, particularly along major road networks, to mitigate rising land surface temperatures.	

Urban Effects on Land Surface Temperature in Davao City, Philippines	Myla Ross Tinoy, A. U. Novero, K.P. C. Landicho, Alvin Baloloy, Ariel Blanco	**Abstract:** This study produced spatiotemporal hot and cold spot occurrence maps for Davao City for the period 1994-2019 using land surface temperature (LST) images. Urban heat is theorized to have been affected by some, if not all, of the following impact factors: air pollutant concentrations/particulate matter (PM10), vegetation “abundance” (using EVI), building “density” (NDBI), albedo, topography, and population density. A mobile traverse sampling was performed in the morning and afternoon of 15 April 2019 to measure PM10 in the city’s identified hot spots. The remaining factors were generated from imagery (i.e., Landsat 8, Synthetic Aperture Radar) and obtained from the Philippine Statistics Authority. These factors were analyzed against the LST which was obtained through Project GUHeat’s methodology. The relationships between the factors and LST were studied through multiple and quantile regression models (MRM & QRM). Results showed that variable PM10 does not have any significance in the MRM. Meanwhile, QRM were fitted to different quantile values, namely: 10th, 25th, 50th, 75th, and 90th. It is only at the 90th quantile where all the independent variables are good predictors for the LST. Albedo is the most important predictor for the LST at 10th quantile whereas Elev for the 25th quantile. However, when LST is at the 50th, 75th, and 90th quantiles NDBI is the most significant variable at predicting LST. Reliable spatiotemporal assessment and modelling of surface temperature are essential for urban planning and management to formulate sustainable strategies for the welfare of people and environment.	**Conclusions and Recommednations:** This study investigated the spatiotemporal distribution of LSTand its quantitative relationships with atmospheric,demographic, and remote sensing-derived parameters in DavaoCity. The results revealed that occupants of the main urbancenter (Districts of Poblacion, Talomo, Agdao, and Buhangin)and residents from coastal barangays of Bunawan and TorilDistricts experienced the warmest temperatures in Davao Cityfor the period 1994-2019. Multiple regression analysis showedthat the air pollutant variable (PM10) exhibited no significance,hence was dropped in the model. Albedo has greater impact inthe MRM followed by NDBI, Elev, Pop_Dens, and EVI. Quantile regression analysis further revealed that at the 90thquantile (31.87 °C – 33.42 °C) where all impact factors are goodpredictors, the presence of buildings in the city (NDBI) showedthe biggest impact for the LST. The result of these two regressionmethods indicated that the models are reliable enough to expressLST of Davao City, but also suggested that other impact factorsare more significant in order to effectively understand thevariations of LST. This study recommends that higher spatialand temporal resolution of satellite imageries should beemployed to continuously improve the HS and CS analyses.Moreover, further research should be done by replacing otherimpact factors in the model and try to include surface moisture,distance to water, solar insolation, and sky view factor. Lastly,to evaluate and compare the urban expansion of other developedareas in Mindanao such as Digos City, Panabo City, and TagumCity performing the same procedure of this study is suggested.

Assessment of Intra-Urban Heat Island in a Densely Populated City Using Remote Sensing: A Case Study for Manila City	Mark Angelo Purio, Tetsunobu Yoshitake, and Mengu Cho	**Abstract:** Changes in the environment occur in cities due to increased urbanization and population growth. Sustainable Development Goal (SDG) 11 is intrinsically linked to the environment, one facet of which is the need for universal access to secure, inclusive, and accessible green and public places. As urban heat islands (UHI) have the potential to negatively influence cities and their residents, existing resources and data must be used to identify and quantify these effects. To address this, we present the use of satellite-derived (2013–2022) and meteorological data (2014–2020) to assess intra-urban heat islands in Manila City, Philippines. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common Land-Use Land Cover indicators (Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Normalized Difference Built-up Index (NDBI)) and Land Surface Temperature (LST), (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI are established.	**Conclusions:** This study presents the use of satellite-derived data and meteorological data to assess the presence of an intra-urban heat island in Manila City, Philippines. To address SDG 11 and provide better insights to make cities and human settlements inclusive, safe resilient, and sustainable in terms of UHI, different assessment methods were used and established. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common LULC indicators (NDVI, NDBI, and NDWI) to LST, (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model. Finally, the areas that need intervention are compared to the affected population, and suggestions to enhance the thermal characteristics of the city and mitigate the effects of UHI were established. Results show that there exists a clear difference between cold and warm areas within Manila City. Overall, residential areas, asphalted and concrete roads and walkways, and some commercial establishments and buildings exhibit higher surface temperatures compared to areas with vegetation and near bodies of water. Based on the results, mitigation strategies applicable to Manila City were proposed to improve the areas which need intervention.

Land-use/land cover change and land surface temperature in Metropolitan Manila, Philippines using landsat imagery	King Joshua Almadrones-Reyes & Nikki Heherson A. Dagamac	**Abstract:** Remote sensing technology has been widely used to address the issue of urbanization for monitoring. In Metropolitan Manila, Philippines, there is a decline in vegetation over the last decade. This leads to the excess heat in the surface temperature and a phenomenon known as Urban Heat Island. In this study, the land surface temperature (LST), land-use/land cover changes (LULC), normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI) in 2001 and 2019 were retrieved using Landsat 7 Enhanced Thematic Mapper + (ETM +). Our results showed the following: (i) an increase in the mean temperature of 4.02 °C from 2001 to 2019 in MM, (ii) rise in NDBI and a decline in NDVI values is correlated with an increase in LST over the last years in MM; (iii) LULC maps affirmed the highest occupancy of built-up areas caused by possible urban sprawl explaining the increase in warming in MM. This study provides added evidence that urbanization activities in Metro Manila leads to a warming effect that drives discomfort to many average Filipinos. Hence, environmental policies that would implement proper urban design utilizing more green spaces is recommended for local government units in MM to support.	

Benchmarking Landsat-8 Collection 2 Level-2 Land Surface Temperature Accuracy Using SURFRAD Stations: Effects of Seasonality and Atmospheric Water Vapor	Almustafa AbdElkader Ayek , Mohannad Ali Loho , Nasser Ibrahem , Afnan Abdullah Alturki ,Youssef M. Youssef, and Mayada Abdelkader Abdelaziz	**Abstract:** Land Surface Temperature (LST) is essential for climate monitoring, drought assessment, and urban heat analysis. Despite its importance, the Landsat-8 Collection 2 Level-2 (C2L2) LST product has not been rigorously validated using ground measurements—a critical gap this study addresses. We present the first comprehensive accuracy assessment using 382 coincident satellite–ground observations collected from seven Surface Radiation Budget Network (SURFRAD) stations distributed across diverse climatic regions of the United States during the period 2023–2025. The validation results indicate strong overall agreement between satellite-derived and ground-measured temperatures, yielding an RMSE of 4.20 °C, a coefficient of determination (R2) of 0.91, and a Pearson correlation coefficient (r) of 0.98. These statistics demonstrate the high reliability of the C2L2 LST product across a wide range of environmental conditions. Nevertheless, a systematic warm bias of 1.75 °C was observed, indicating a tendency toward temperature overestimation. Model performance exhibited pronounced seasonal variability. The highest accuracy was achieved during winter conditions (RMSE = 2.17 °C; r = 0.99), whereas performance declined considerably during summer months (RMSE = 5.84 °C; r = 0.91). Analysis of atmospheric water vapor content revealed significant associations with retrieval errors at high-elevation and arid locations, particularly at FPK (r = 0.78) and DRA (r = 0.75), based on 106 matched observations. These relationships provide important insight into the atmospheric factors contributing to seasonal variations in retrieval accuracy. Temperature-dependent analyses further demonstrated that retrieval uncertainty increases with surface temperature. Performance progressively deteriorated from cooler to warmer thermal regimes, with RMSE values increasing from approximately 2.05 °C for temperatures below 20 °C to 5.71 °C for temperatures exceeding 40 °C. Spatial evaluation also revealed substantial differences among stations. Relatively homogeneous, low-elevation sites exhibited superior performance (GWN: RMSE = 2.60 ° C; SXF: RMSE = 2.55 °C), whereas stations located in mountainous or topographically complex environments showed reduced accuracy (TBL: RMSE = 5.14 °C; FPK: RMSE = 5.62 °C). These outcomes emphasize the influence of terrain complexity and atmospheric heterogeneity on LST retrieval performance. Overall, this study establishes the first comprehensive benchmark for evaluating the reliability of Landsat-8 C2L2 LST products. The results provide valuable guidance for their application in climate research, precision agriculture, hydrological modeling, and environmental monitoring. Furthermore, the findings identify specific environmental conditions requiring enhanced validation efforts and suggest opportunities for future algorithm refinement through improved atmospheric correction procedures and more	

Use of normalized difference built-up index in automatically mapping urban areas from TM imagery	Y. Zha,J. Gao &S. Ni	**Abstract:** Remotely sensed imagery is ideally used to monitor and detect land cover changes that occur frequently in urban and peri-urban areas as a consequence of incessant urbanization. It is a lengthy process to convert satellite imagery into land cover map using the existing methods of manual interpretation and parametric image classification digitally. In this paper we propose a new method based on Normalized Difference Built-up Index (NDBI) to automate the process of mapping built-up areas. It takes advantage of the unique spectral response of built-up areas and other land covers. Built-up areas are effectively mapped through arithmetic manipulation of re-coded Normalized Difference Vegetation Index (NDVI) and NDBI images derived from TM imagery. The devised NDBI method was applied to map urban land in the city of Nanjing, eastern China. The mapped results at an accuracy of 92.6% indicate that it can be used to fulfil the mapping objective reliably. Compared with the maximum likelihood classification method, the proposed NDBI is able to serve as a worthwhile alternative for quickly and objectively mapping built-up areas.	

Estimation of land surface temperature–vegetation abundance relationship for urban heat island studies	Qihao Weng, Dengsheng Lu, Jacquelyn Schubring	Abstract: Remote sensing of urban heat islands (UHIs) has traditionally used the Normalized Difference Vegetation Index (NDVI) as the indicator of vegetation abundance to estimate the land surface temperature (LST)–vegetation relationship. This study investigates the applicability of vegetation fraction derived from a spectral mixture model as an alternative indicator of vegetation abundance. This is based on examination of a Landsat Enhanced Thematic Mapper Plus (ETM+) image of Indianapolis City, IN, USA, acquired on June 22, 2002. The transformed ETM+ image was unmixed into three fraction images (green vegetation, dry soil, and shade) with a constrained least-square solution. These fraction images were then used for land cover classification based on a hybrid classification procedure that combined maximum likelihood and decision tree algorithms. Results demonstrate that LST possessed a slightly stronger negative correlation with the unmixed vegetation fraction than with NDVI for all land cover types across the spatial resolution (30 to 960 m). Correlations reached their strongest at the 120-m resolution, which is believed to be the operational scale of LST, NDVI, and vegetation fraction images. Fractal analysis of image texture shows that the complexity of these images increased initially with pixel aggregation and peaked around 120 m, but decreased with further aggregation. The spatial variability of texture in LST was positively correlated with those in NDVI and in vegetation fraction. The interplay between thermal and vegetation dynamics in the context of different land cover types leads to the variations in spectral radiance and texture in LST. These variations are also present in the other imagery, and are responsible for the spatial patterns of urban heat islands. It is suggested that the areal measure of vegetation abundance by unmixed vegetation fraction has a more direct correspondence with the radiative, thermal, and moisture properties of the Earth's surface that determine LST.	

Google Earth Engine: Planetary-scale geospatial analysis for everyone	Noel Gorelick, Matt Hancher, Mike Dixon, Simon Ilyushchenko, David Thau , Rebecca Moore	**Abstract:** Google Earth Engine is a cloud-based platform for planetary-scale geospatial analysis that brings Google's massive computational capabilities to bear on a variety of high-impact societal issues including deforestation, drought, disaster, disease, food security, water management, climate monitoring and environmental protection. It is unique in the field as an integrated platform designed to empower not only traditional remote sensing scientists, but also a much wider audience that lacks the technical capacity needed to utilize traditional supercomputers or large-scale commodity cloud computing resources.	**Extra Info:** **Median Compositing Rationale & Methodology** "As an illustrative example, a user might wish to compute the difference between two seasonal composites, to highlight the changes due to phenology or snow-cover. A simplistic example of this could be expressed using the Earth Engine client library as the subtraction of two composite images (Listing 1). This code creates two filtered collections, one of all Landsat 8 images for November, December, and January, and a second of all Landsat 8 images from June, July, and August. A temporal median value is computed for each band in each collection (to minimize the effects of cloud and cloud shadows), and the resulting composites are subtracted to compute the change in values. This computational description is represented by the DAG in Fig. 3." **Cloud Screening & QA Masking Applications** "Hansen et al. (2013) characterized forest extent, loss, and gain from 2000 to 2012 using decision trees generated from an extensive set of training data and a deep stack of metrics computed from large collections of Landsat scenes. Filtering operations supported by the data catalog reduced the 1.3 M Landsat scenes available at the time to 654,178 growing-season scenes from the study period. These images were then screened for clouds, cloud shadows, and water, and converted from raw Landsat digital numbers to normalized top of atmosphere reflectance. All necessary data access, format conversion, cartographic reprojection, and resampling were handled automatically by the system. Operations in the API were used to compute input metrics, such as per-band percentile values and linear regressions of reflectance values versus image date." "They used Earth Engine to select, on a per-pixel basis, the best Landsat scenes for the early and late periods, by first calculating reflectance of the Landsat scenes using LEDAPS (Masek et al., 2006), automatically removing cloudy scenes using Earth Engine's SimpleCloudScore function, computing GCVI values, and finally selecting the scenes with the highest GCVI. Once the best pair of Landsat scenes was determined for

Google Earth Engine Cloud Computing Platform for Remote Sensing Big Data Applications: A Comprehensive Review	Meisam Amani; Arsalan Ghorbanian; Seyed Ali Ahmadi; Mohammad Kakooei; Armin Moghimi; S. Mohammad Mirmazloumi	**Abstract:** Remote sensing (RS) systems have been collecting massive volumes of datasets for decades, managing and analyzing of which are not practical using common software packages and desktop computing resources. In this regard, Google has developed a cloud computing platform, called Google Earth Engine (GEE), to effectively address the challenges of big data analysis. In particular, this platform facilitates processing big geo data over large areas and monitoring the environment for long periods of time. Although this platform was launched in 2010 and has proved its high potential for different applications, it has not been fully investigated and utilized for RS applications until recent years. It was observed that Landsat and Sentinel datasets were extensively utilized by GEE users. Moreover, supervised machine learning algorithms, such as Random Forest, were more widely applied to image classification tasks. GEE has also been employed in a broad range of applications, such as Land Cover/land Use classification, hydrology, urban planning, natural disaster, climate analyses, and image processing. It was generally observed that the number of GEE publications have significantly increased during the past few years, and it is expected that GEE will be utilized by more users from different fields to resolve their big data processing challenges.	**Extra Info:** "Landsat satellites have optical sensors, the images of which may be obscured by clouds. Therefore, temporal cloud detecting, masking, and removing are essential preprocessing steps in different applications, such as image classifications using multitemporal imagery [12]."

Google Earth Engine Open-Source Code for Land Surface Temperature Estimation from the Landsat Series	Sofia L. Ermida, Patrícia Soares, Vasco Mantas, Frank-M. Göttsche, Isabel F. Trigo	**Abstract:** Land Surface Temperature (LST) is increasingly important for various studies assessing land surface conditions, e.g., studies of urban climate, evapotranspiration, and vegetation stress. The Landsat series of satellites have the potential to provide LST estimates at a high spatial resolution, which is particularly appropriate for local or small-scale studies. Numerous studies have proposed LST retrieval algorithms for the Landsat series, and some datasets are available online. However, those datasets generally require the users to be able to handle large volumes of data. Google Earth Engine (GEE) is an online platform created to allow remote sensing users to easily perform big data analyses without increasing the demand for local computing resources. However, high spatial resolution LST datasets are currently not available in GEE. Here we provide a code repository that allows computing LSTs from Landsat 4, 5, 7, and 8 within GEE. The code may be used freely by users for computing Landsat LST as part of any analysis within GEE.	**Extra Info:** On justifying Landsat LST for urban heat mapping: "Land Surface Temperature (LST) is an important component of the Earth's energy budget, closely linked to the partitioning between sensible and latent heat fluxes. As such, high-resolution satellite-derived LST is increasingly used in various applications related to the assessment of land surface conditions, including mapping the urban extent and the intensity of urban micro-climates, estimating high-resolution evapotranspiration for the management of water resources and assessing vegetation stress [1-15]." On justifying temporal consistency and comparability (using Tier 1 data for time-series): "The data are organized into Tiers based on their quality. Landsat scenes with the highest available data quality are placed into Tier 1, being suitable for time-series analysis, while the remainder are assigned to Tier 2 [42]. The results shown in this work are entirely based on Tier 1 data, but the code may equally be applied to Tier 2 data." On justifying cloud-screening procedures: "Cloud coverage information, including cloud shadowing, can be retrieved from the quality assessment band (BQA), which is also made available by the USGS through GEE." "Based on this information, the main module, Landsat_LST, loads the respective collections of TOA brightness temperatures (BT) and Surface Reflectance (SR). A cloud mask is applied to both using the quality information bands (module cloudmask)."

Enhancing the spatial and temporal resolution of satellite-derived land surface temperature in urban environments: A systematic literature review	Robert Andriambololonahari soamalala , Petra Helmholz ,Ivana Ivánová, Eriita Jones, Susannah Soon , Dimitri Bulatov , Yongze Song	**Abstract:** The urban heat island effect significantly challenges human health, energy consumption, and overall urban living conditions. Accurate land surface temperature (LST) measurement at fine spatial and temporal scales is essential to understand urban thermal dynamics and develop effective mitigation strategies. However, current satellite sensors offer a trade-off between spatial and temporal resolutions. This review critically examines recent techniques for enhancing the resolutions of satellite-derived LST, focusing on urban applications. A comprehensive search was conducted with Scopus and Web of Science databases using keywords related to LST resolution enhancement. After applying criteria to remove irrelevant and duplicate studies, 60 papers were selected for in-depth analysis. Our findings indicate that LST data from polar and geostationary orbit satellite thermal infrared (TIR) sensors are mainly used for resolution enhancements. The identified enhancement methods fall into three categories: spatial, temporal, and combined spatial and temporal approaches. Existing techniques achieve spatial and temporal resolutions, respectively, 10 m and 15 min. High-resolution LST data are valuable for urban planners and policymakers to assess public health risks, energy use, insurance considerations, and emergency response planning. This review provides insights into current LST resolution enhancement methods and their practical implications in addressing urban heat challenges.	Extra Info: "The urban environment consists of a complex heterogeneous surface that significantly influences the formation of UHI and cannot be adequately described using traditional remote sensing techniques based on land cover classification." "TIR data usually exhibits a coarser spatial resolution compared to visible and near-infrared (VNIR) data obtained from the same platform because of the difference in energy emission and instrumental limitations... This coarse resolution may lead to a blending effect in areas where one TIR pixel may cover an area that is bigger than the thermal features within it." "In terms of the variety of land cover, LST at a regional level is insufficient to reflect the variations of LST within cities. To study urban climate, better spatial resolution LST data is required to distinguish between urban characteristics such as streets, buildings, and roads."